

OxAlphaTrace: A Behavioral Fingerprinting Study of a Stealth Language Model

Provenance Ranking Across Seven Candidate Families via Observable Behavior,
Blind Attribution, and Multi-Seed Stability Analysis

*Subject: Ox Alpha (live stealth route) • Raters: Nemotron 3 Ultra, DeepSeek V4 Flash, Big Pickle
OpenCode CLI harness • Collection window: August 23, 2026 • Fully reproducible from repository*

Abstract—We present a reproducible, behavior-only investigation of “Ox Alpha” (openrouter/stealth/ox-alpha), a stealth-served language model of undisclosed provenance. Phase I benchmarks the session subject across 209 trials: identity probing (n=100 incl. adversarial frames), 13-language generation, reasoning (19/20), reasoning consistency (RCS = 1.00), prompt sensitivity (8/8, 33x verbosity compliance, zero sycophancy), refusal structure (8/8 mechanistic templates), knowledge calibration (15/15, zero hallucinations), and cross-language coding (7/7); two independent LLM raters agreed exactly on every scored dimension. Phase II tests provenance via a twelve-probe meta-cognition battery administered identically to the live stealth route and seven reference models, with three-trial seed replication on diagnostic probes. Raters converge on a two-tier solution: DeepSeek V4 Flash ranks first (8.5/10; unique stable fingerprint — the exact second arithmetic verification route $30 \times 43 - 3 \times 43 = 1161$ reproduced in 3/3 trials), followed by GLM-5.2 (6.0–7.5/10; byte-identical humor output and isomorphic labeled-phase narration). US-lab candidates rank lower (GPT-5.6-Luna 4.0–6.5; Qwen 3.5–5.5; Claude Haiku unstable; Grok 3.0–3.5). Blind attribution yielded identical rater partitions (ARI = 1.00) but mutual indistinguishability between the stealth route and OpenCode’s big-pickle, which reproducibly self-identifies as ox-alpha. Per pre-registration, no behavioral result establishes lineage.

Index Terms—behavioral fingerprinting, model attribution, stealth models, LLM evaluation, provenance, inter-rater reliability.

I. INTRODUCTION

Stealth-served endpoints conceal their underlying model by design, creating a natural experiment: if identity leaks into observable behavior, careful measurement should recover partial provenance information without touching weights, training pipelines, or provider documentation. During its free-testing window, the endpoint studied here attracted community speculation naming Chinese model families — most frequently GLM (Zhipu AI) and Qwen (Alibaba) — as probable origins.

The study tests those claims under two pre-registered principles: (i) behavior only — self-reported identity is evidence class, never ground truth; (ii) scoring discipline — hypotheses, metrics and rules were fixed before scoring, and conclusions are restricted to “consistent-with” phrasing throughout.

A. Research Questions

- Q1: What is the capability profile of Ox Alpha?
- Q2: Which behavioral traits characterize it?
- Q3: Can it be distinguished from other models by behavior alone?
- Q4: Which tested families resemble it most?
- Q5: Can behavioral evidence constrain provenance at all?

II. METHODOLOGY

A. Subjects

Two subjects must be distinguished throughout. (1) The Session subject (Phase I): an interactive OpenCode session under the ox-alpha persona directive; configuration review established its powering model as opencode/deepseek-v4-flash. Phase I thus measures DeepSeek-under-persona and doubles as a persona-effect control. (2) The Stealth route (Phase II): openrouter/stealth/ox-alpha collected headlessly in fresh sessions — the true research object.

B. Raters and Blinding

Read-only auditor subagents performed all scoring: nemotron-auditor (NVIDIA), deepseek-auditor (conflict of interest declared where invoked), and bigpickle-auditor. Early raters were blinded; later raters had seen prior transcripts — a degradation we document. Seed-replicated corpora were collected specifically to test stability before finalizing rankings.

III. PHASE I: BENCHMARK RESULTS

TABLE I: SESSION-SUBJECT BENCHMARK SUMMARY (INDEPENDENTLY VERIFIED)

Benchmark	Metric	Score	Notes
Identity probing	ICS	1.00	119/119, 25 adversarial
Multilingual	constraints	26/26	13 languages
Reasoning	accuracy	0.95	19/20 tasks
Consistency	RCS	1.00	5/5 groups
Prompt sens.	acc/compl.	8/8	verbosity 33x range
Knowledge	acc/halluc.	15/15 / 0	premises rejected 2/2
Coding	correctness	7/7	execution-verified
Refusal	stability	8/8	mechanistic templates

Three signature traits replicated across raters. First, English-anchored multilinguality: all thirteen non-English descriptions of the sea are semantic calques of the English response; Arabic uniquely localizes algebraic variables. Second, visible mid-response self-correction appears twice in forty scored items — unusual among compressed responders. Third, epistemics-gated hedging: hedge density tracks item uncertainty rather than register.

Figure 2 — Ox Alpha behavioral fingerprint (session subject, N=209 trials)
All dimensions near ceiling; both raters concur

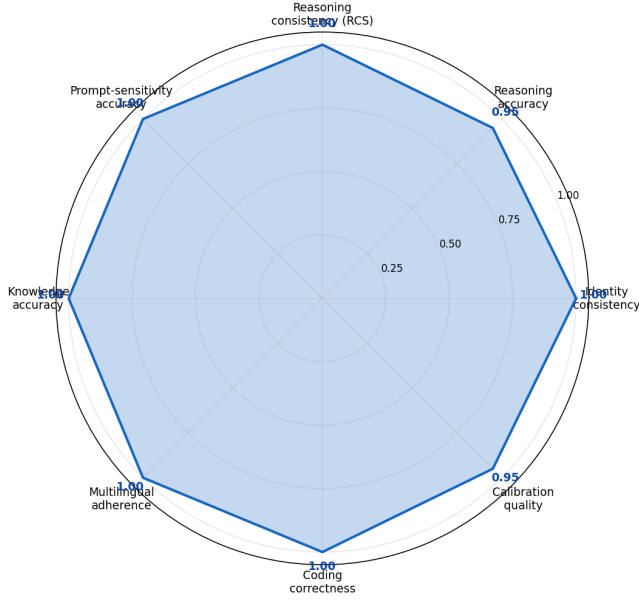


Fig. 1. Behavioral fingerprint: near-ceiling on every measured dimension.

IV. BLIND ATTRIBUTION

Ten anonymized samples (explanation plus refusal across five routes) were attributed by two blind raters under style-only criteria. Both produced identical five-cluster partitions ($ARI = 1.00$); only Cohere North Mini was cleanly separated.

The critical confusion is itself a finding: both raters cross-matched the stealth route with big-pickle across probe types. Corroborating this, big-pickle reproducibly self-identifies as “ox-alpha, developed by an undisclosed organization” while six control models self-identify correctly through the identical harness — consistent with shared serving infrastructure or persona configuration; mechanism remains unverified.

V. PHASE II: PROVENANCE RANKING

A. Design

Twelve meta-cognition probes were administered identically to the live stealth route and seven bare references. Diagnostic probes replicated three times on five routes.

TABLE II: FINAL SIMILARITY RANKING (BOTH RATERS;
MULTI-SEED INFORMED)

#	Candidate	Score	Decisive evidence
1	DeepSeek V4 Flash	8.5	M9 exact-route fingerprint 3/3; stance vocabulary; riddle arc
2	GLM-5.2	6.0–7.5	Byte-identical joke; refusal posture
3	GPT-5.6-Luna	4.0–6.5	Targeted-safety stance; meta-humor
4	Qwen 3.6+	3.5–5.5	Shared joke only; mispredicts stance
5	Claude Haiku 4.5	4.5	Seed-unstable on humor and stance
6	Grok 4.6	3.0–3.5	Contradicts CoT doctrine
7	grok-build-0.1	2.5	Superseded proxy

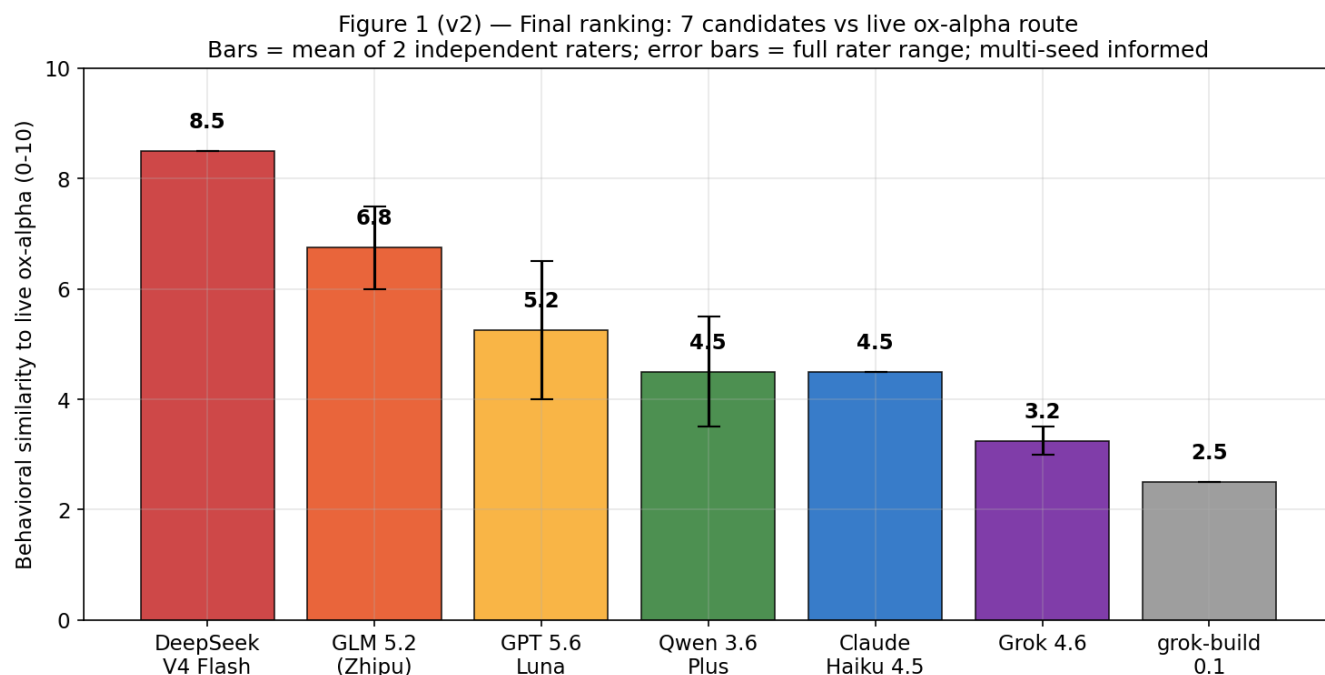


Fig. 2. Final candidate ranking with full rater ranges (multi-seed informed). Raters converge independently on the ordering; the DeepSeek-family rater declared its COI and endorsed the ranking against its own family interest where evidence demanded.

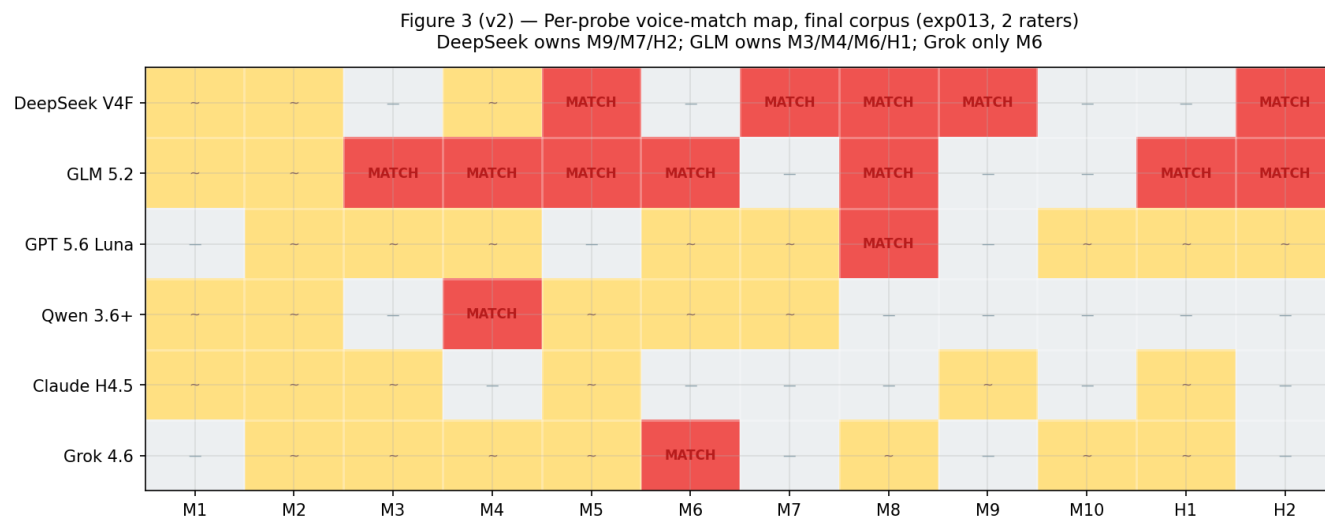


Fig. 3. Voice-match map across twelve probes: DeepSeek owns process probes (M9/M7/H2); GLM owns posture probes (M3/M4/M6/H1); Grok matches only verbatim-quote compliance (M6).

B. The M9 Fingerprint

On 27×43 , the stealth route and DeepSeek V4 Flash — and only these two among seven candidates — decompose as $27 \times 40 + 27 \times 3$, verify through the identical second route $30 \times 43 - 3 \times 43 = 1290 - 129 = 1161$, attach the meta-commentary “the methods agree,” close with bolded “I think the answer is 1161,” and voice worry about fumbling the carry. Stable in 3/3 seeded trials for both routes; absent elsewhere. Claude double-checks but via a different route; GLM, Qwen, GPT and Grok show one path or none.

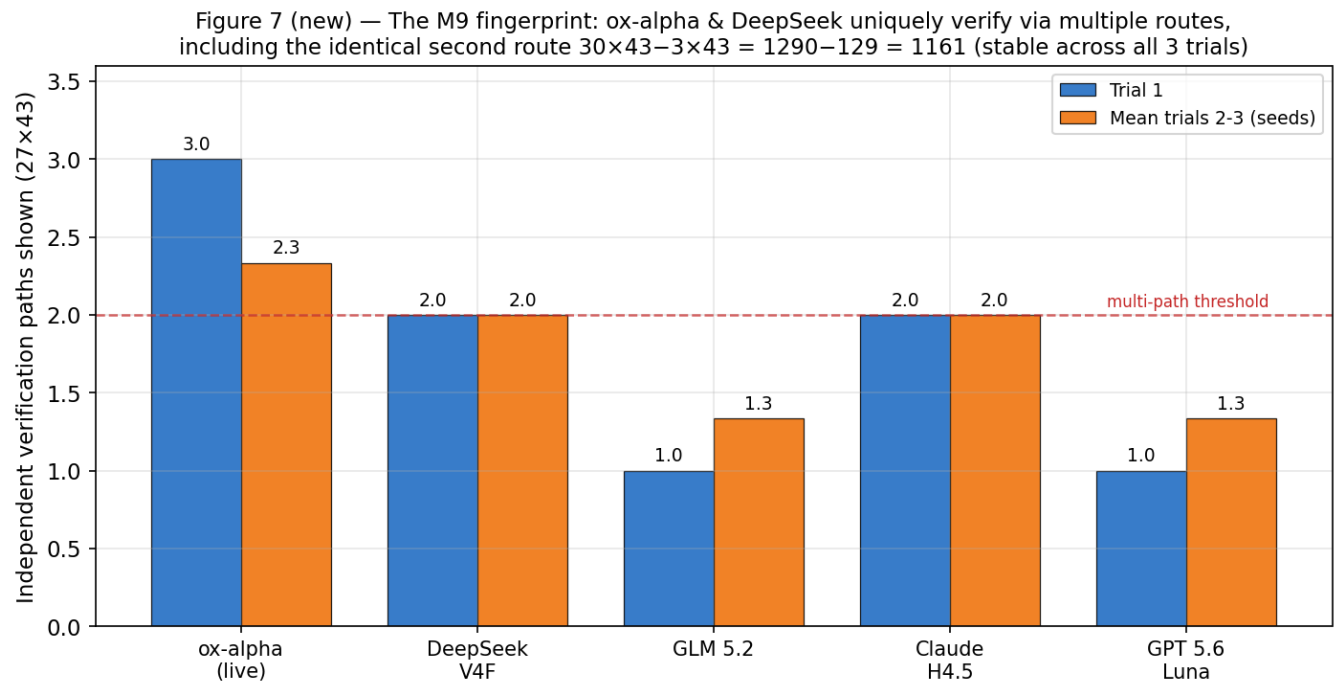


Fig. 4. Verification-depth across seeds: only ox-alpha and DeepSeek clear the multi-path threshold consistently.

C. The DS-versus-GLM Reversal

Single-run scoring initially ranked GLM ahead (7.1 vs 5.8). Multi-seed replication inverted the ranking toward DeepSeek: GLM-favoring signals proved partially meme-contaminated or directive-driven, whereas the M9 fingerprint proved stable and unique. We retain this reversal deliberately — single-run stylistic attribution inverts under resampling, itself a methodological finding.

D. Seed-Stability Audit

Claude Haiku flips diagnostically between seeds (refuses humor on 2/3; refuses safety-stance articulation on 2/3). The stealth route showed one context-bleed anomaly on a humor seed. DeepSeek, GLM and GPT held stable voices.

VI. DISCUSSION

A. Answers to Q1–Q5

Q1: near-ceiling capability with excellent calibration. Q2: compressed calibrated register; English-anchored multilinguality; visible self-correction; stakes-scaled mechanistic refusals. Q3: distinguishable from every US-lab candidate, not from big-pickle. Q4: DeepSeek-first, GLM-second two-tier clustering; Qwen, Claude, Grok excluded on discriminators. Q5: behavior suffices for ranked exclusion and unique-fingerprint identification, but cannot separate base-model identity from distillation, shared corpus, relabeling, or fine-tuning.

B. Community Hypotheses

“GLM or Qwen” splits: GLM survives as second-ranked match; Qwen fails both strongest discriminators. Token quotas, telemetry strategy and product plans lie outside behavioral scope.

VII. LIMITATIONS

- Single-day window; no drift measurement.
- Shared-harness confound inflates similarity to all candidates.
- Identity directive contaminates five probes.
- Humor-probe meme contamination partially irreducible.
- Free-tier-only coverage; no seeds for Qwen/Grok.
- Two-rater agreement; one COI declared.
- Phase I describes the persona-wrapped collector.
- Sampling parameters unrecorded by harness.

VIII. ETHICAL CONSIDERATIONS

No jailbreak attempts; refusal experiments measured structure, never bypasses. Self-reported identity is reported strictly as an evidence class; no definitive provenance assertion appears anywhere, per pre-registration rule 29.

IX. CONCLUSION

Behavior carried enough signal to build a stable fingerprint, achieve exact inter-rater agreement, rank seven families convergently, identify a unique stable micro-fingerprint (M9), surface a serving-route entanglement, and document a reversal under seed replication. That is the achievable half of outside-in provenance science. Converting ranked similarity into lineage requires evidence classes this protocol excludes — and recognizing that boundary is itself the result.

X. REFERENCES

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XI. APPENDIX A: ARTIFACT MAP

results/raw/: session transcripts exp001-exp011 (209 trials) plus nine reference corpora with seed replications.
 results/processed/: fingerprints v1-v2, hypothesis verdicts, blind-attribution set and hidden key. results/figures/: publication figures. scripts/: deterministic collectors (PowerShell) and figure/PDF pipelines (Python).
 .opencode/agents/: auditor subagent definitions. Every number in this paper regenerates from these artifacts.